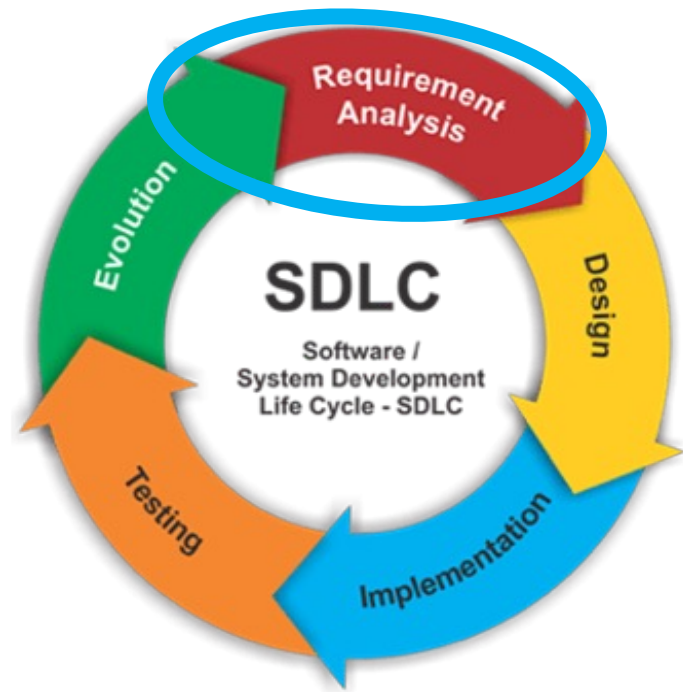


Chapter 3: User Requirements Gathering and Analysis - 1



User Requirements Gathering and Analysis in SDLC

- The first step of any SDLC
- Define the project's requirements.
- During this phase, all the relevant information is collected to develop a product as per expectation.



Stakeholders

Stakeholders are those with an interest in the project's outcome.

Stakeholders can influence towards the success or failure of the software project.



Requirements

Requirement specifies what an intended service or product should perform.

Product requirement provides all functions, features and behaviors where the system must possess so that it will be efficient to use.

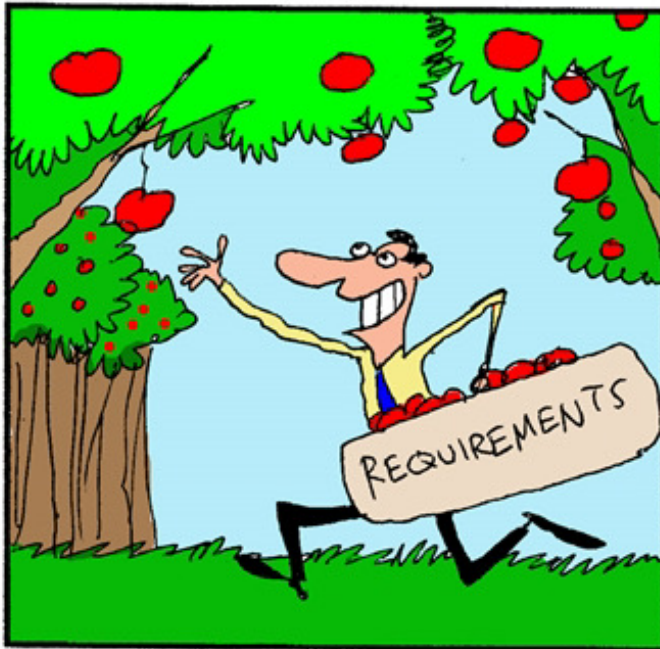
Requirements engineering is about establishing and managing requirements.

Requirements elicitation is the practice of gathering, researching and verification of requirements of a system

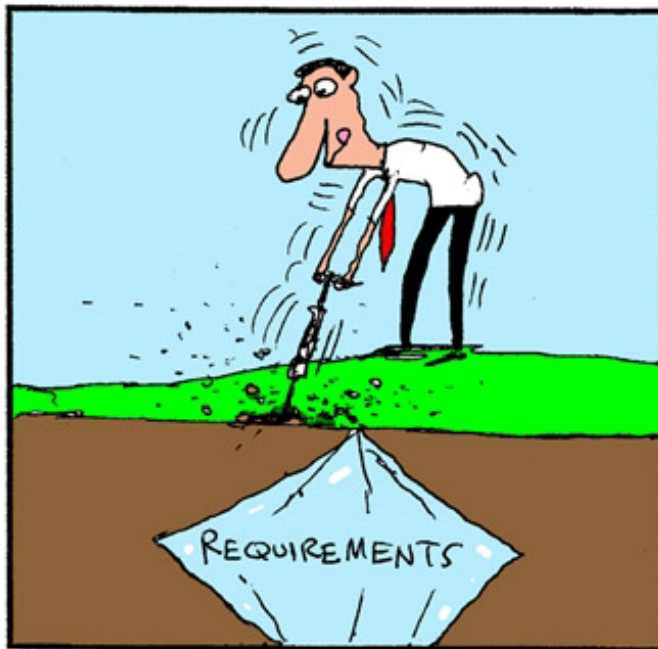
Requirements Gathering

Cartoon by Ryan Hewitt

How stakeholders think requirement gathering works.



How requirement gathering really works.



Requirements to be Gathered

- Business Rule
- Business Requirements
- External Interface Requirements
- Functional Requirements
- Non-Functional Requirements
- Constraints
- etc.

Business Rules and Requirements

Business Rules

- Provides the criteria, conditions and exceptions on a specific action.

Business Requirements

- Most crucial part at the level of SDLC.
- Provides brief set of business functionalities that the system needs to meet to be successful.
- Define a project's business need and criteria of its success
- Does not define technical details such as the type of technology implemented in the system.
- Shows no such detail as to how the system will implement this requirement, but rather what the system must do concerning the business.

Business Requirements are used to implement Business Rules.

Example of Business Rules and Requirements

Business Rule: You must be 21 or older to view the alcohol product website.

(Option 1) Business Requirement: The user shall provide a birthdate as proof of their age.

(Option 2) Business Requirement: The user shall provide a valid drivers license as proof of their age.

Example of Business Rules

Business Rules are typically expressed using formal logic qualifiers such as: "IF-THEN", "IF-ELSE", "ONLY IF", "WHEN"

For an online order system

- Customer must have a valid email address

For an ATM withdrawal transaction

- Customer must enter correct pin number

For a company's purchasing approval system, the following rules are applied:

		Rules		
		1	2	3
Input	Invoice Amount Range	≤ \$10000	\$10000 to \$50000	≥ \$50000
Output	Approver	Josephine Sully	Dewey McCormick	Linda Payne

External Interfaces Requirements

- Provide system integration with other components.
- Identify and document the interfaces to other systems and external entities within the project scope.
- Belong to functional requirements that are important for systems.

Type of External Interfaces Requirements

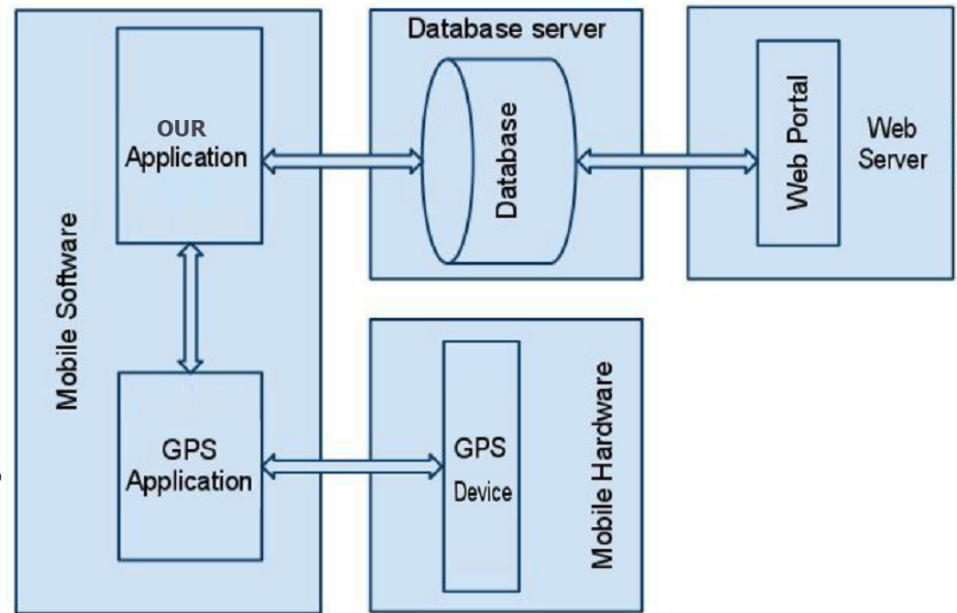
- User interface
 - graphical user interface
 - command line interface
 - voice-enabled user interface
- Hardware interface
 - what devices the software is created for
- Software interface
 - database management system
- Communications interface
 - wireless or wired technologies used to connect devices to one another

Partial Example of External Interfaces

A mobile application requires both Internet and GPS connection to fetch and display results.

All system information is maintained in a database, which is located on a web-server.

The mobile application interacts with the GPS-Navigator software, which is required to be already installed on the user's mobile phone.



Functional Requirements & Non-Functional Requirements

Functional requirements

- Specify what the system should do.
- Usually expressed by users.
- It describes the system services in details.
- Example:
 - Authentication
 - Administrative Functions
 - External Interfaces
 - Business Rules
 - Business Requirements
 - Transaction Corrections

Non-Functional requirements

- Specifies how the system should do it.
- Prepared by technician / software architect / analyst.
- Provide good user experience, efficient and ease to use.
- Also known as quality requirements
- Example:
 - Performance
 - Reliability
 - Usability
 - Security
 - Scalability

Functional Requirements & Non-Functional Requirements

The system will still work if the non-functional requirements are not met.

- A website that takes more than 30 seconds to load will still meet the function functional requirement but fails non-functional requirement

What are Non-functional Requirements and How Do They Work?

<https://youtu.be/fc-5HJPBZMQ>

Constraint Requirement

Constraints can be business or technical in nature.

Defined as restrictions or limitations on possible solutions.

Example of constraints:

- project budget,
- time restrictions,
- manpower limitation,
- technical architecture decisions,
- etc.

Summary

1. Requirements engineering is about establishing and managing requirements.
2. Requirements elicitation is the practice of gathering, researching and verification of requirement of a system
3. Stakeholders can influence towards the success or failure of the software project.

Summary

4. Business requirement

provides brief set of business functionalities that the system needs to meet to be successful.

defines a project's business need and criteria of its success.

is the most crucial part at in SDLC level.

does not define technical details.

5. Product requirement provides all functions, features and behaviors where the system must possess so that it will be efficient to use.

Summary

6. External interface requirements

Provide system integration with other components.

Identify and document the interfaces to other systems and external entities.

7. Type of external interface requirements

- User interface
- Hardware interface
- Software interface
- Communications interface

Summary

8. Functional requirements specify what the system must do and describe system services in detail.

Examples of functional requirements:

- Business rules
- Administrative functions
- Authentication

9. Non-functional requirements specify how the system does it and its operation capabilities and constraints.

Examples of non-functional requirements:

- performance,
- reliability,
- communication between system should be encrypted

References

<https://www.softwaretestinghelp.com/software-development-life-cycle-sdlc/>

<https://www.altexsoft.com/blog/business/functional-and-non-functional-requirements-specification-and-types/>